

# SC4001 Neural Networks and Deep Learning

## Part A: HDB Resale Price Regression

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### Generative AI Declaration

I used **Codex** to assist with the generation of some of the code segments during the development of this assignment. The tool was used primarily to generate initial code templates and to obtain suggestions for implementing specific programming tasks. All AI-generated code was carefully reviewed, modified where necessary, integrated into the overall solution by me, and tested to ensure it met the assignment requirements. I verified the correctness of the generated code through manual inspection, compilation/execution testing, debugging, and comparison with the assignment specifications and relevant course materials. The final submitted work represents my own understanding and effort. I take full responsibility for the accuracy, functionality, and academic integrity of the submitted solution.

**Generative AI Tool Used:** Codex

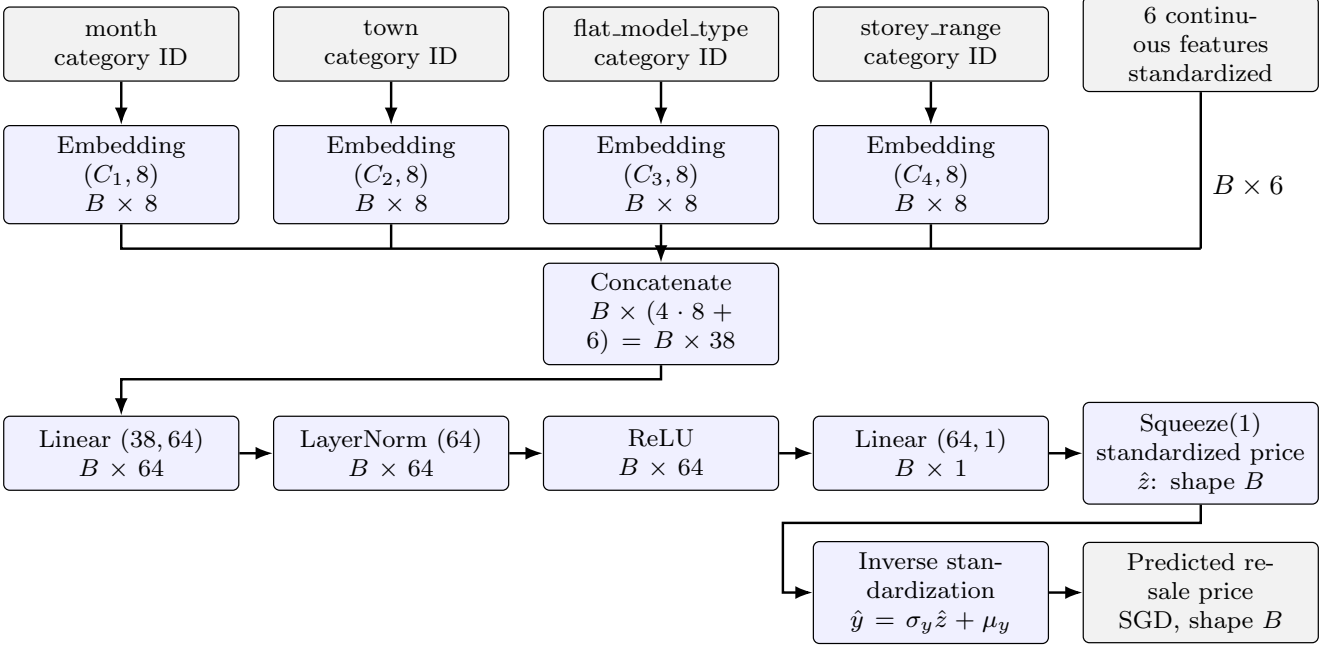
**Purpose of Use:** Programming assistance, README.md generation, and LaTeX formatting

**Verification Method:** Manual code review, testing, debugging, and validation against assignment requirements

# A1

## (a) Block Diagram & Parameter Count

Block Diagram:



Explanation:

- Each categorical embedding has dimension 8.
- The hidden layer has width 64.
- $B$  denotes the batch size.
- $C_j$  counts the known training categories for feature  $j$ , plus one extra category with ID 0 for unknown values.

### Number of trainable parameters:

Each categorical feature  $j$  uses an embedding table with  $C_j$  rows and 8 trainable values per row. We add one extra category for unknown category for each feature, which gives:

$$C_j = \text{number of categories in feature } j + 1.$$

Calculate  $C_j$  for every categorical features:

| Categorical feature | Number of training categories | $C_j$ calculation   |
|---------------------|-------------------------------|---------------------|
| month               | 12                            | $C_1 = 12 + 1 = 13$ |
| town                | 26                            | $C_2 = 26 + 1 = 27$ |
| flat_model_type     | 43                            | $C_3 = 43 + 1 = 44$ |
| storey_range        | 17                            | $C_4 = 17 + 1 = 18$ |

Six continuous features: `dist_to_nearest_stn`, `dist_to_dhoby`, `degree centrality`, `eigenvector centrality`, `remaining_lease_years`, and `floor_area_sqm`.

Each contributes one input value. Concatenating the four 8-dimensional embeddings with these six values gives the total input dimension  $D$ :

$$D = 4 \times 8 + 6 = 38.$$

Layer-by-layer trainable parameter counts:

| Component             | Calculation                    | Trainable parameters |
|-----------------------|--------------------------------|----------------------|
| Four embedding tables | $(13 + 27 + 44 + 18) \times 8$ | 816                  |
| Linear (38, 64)       | $38 \times 64 + 64$            | 2,496                |
| LayerNorm (64)        | $64 + 64$                      | 128                  |
| Linear (64, 1)        | $64 \times 1 + 1$              | 65                   |
| <b>Total</b>          | $816 + 2,496 + 128 + 65$       | <b>3,505</b>         |

Explanation:

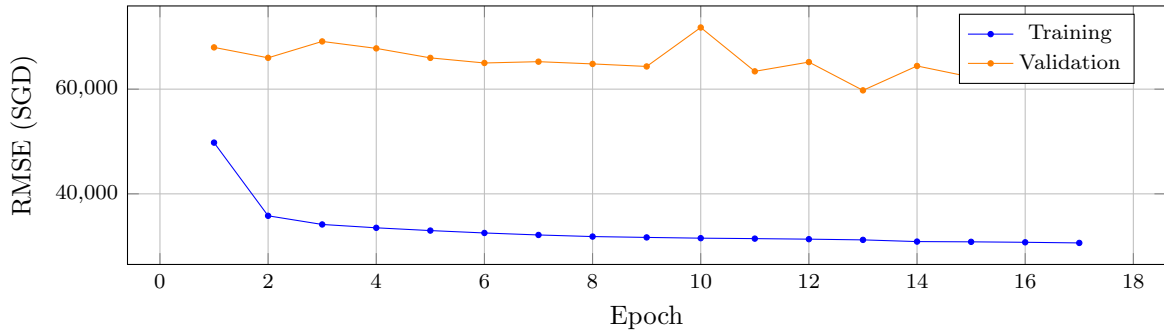
- Each linear layer includes a weight matrix and a bias vector.
- LayerNorm has 64 learnable scale parameters and 64 learnable shift parameters.
- Concatenation, ReLU, squeezing, and inverse target standardization add no trainable parameters.

## (b) Hyperparameter Search

- **Search method:** Ray Tune grid search over 25 combinations: Hidden-layer width  $h \in \{16, 32, 64, 128, 256\}$  and embedding dimension  $d \in \{2, 4, 8, 12, 16\}$  shared by all four categorical features.
- **Data split:** Training on 2017–2020 and validation on 2021.
- **Training settings:** Seed 42; Adam with learning rate 0.01; batch size 512; maximum 20 epochs; early-stopping patience 4 and minimum validation RMSE improvement  $10^{-4}$ .
- **Best result:**  $h = 128, d = 12$ ; validation RMSE 59,766.91 SGD at epoch 13. Training stopped at epoch 17 and epoch-13 weights were retained.

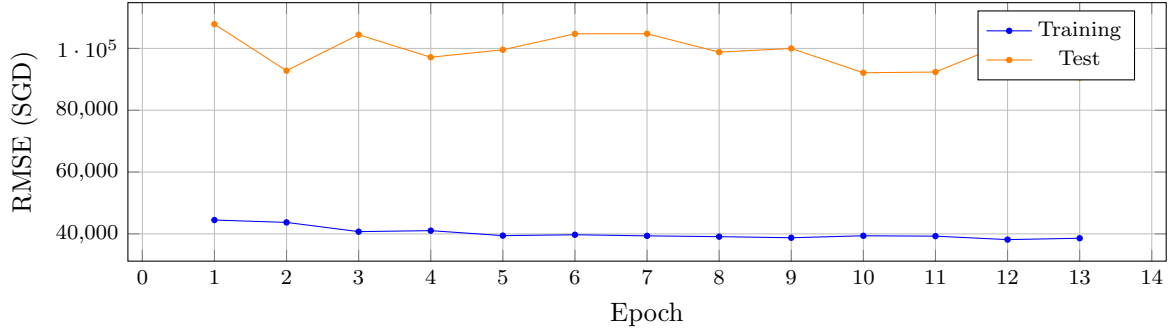
## (c) Selected Configuration

- **Optimal hyperparameters:** Hidden-layer width 128; embedding dimension 12 for each categorical feature.
- **Best validation RMSE:** 59,766.91 SGD at epoch 13.
- **Learning curves:** Training and validation RMSE are shown below.



## (d) Retraining and Test Performance

- **Retraining:** A fresh model with the selected architecture was trained on 2017–2021 (116,427 records), with preprocessing refitted on this combined set. Seed 42 and the A1(b) optimiser settings were retained.
- **Epoch count:** Fixed at 13 using the best validation epoch from A1(c).
- **Evaluation:** Training and testing RMSE were computed after each epoch in SGD. The 2022 test set contained 26,702 records.
- **Final test performance:** RMSE = 90,595.88 SGD;  $R^2 = 0.716868$ .



## (e) Discussion

- **Configuration choice:**  $(h, d) = (128, 12)$  achieved the lowest validation RMSE (59,766.91 SGD), improving on the original (64, 8) configuration (63,422.46 SGD) by 5.76%. The runner-up (256, 2) scored 59,926.48 SGD, only 159.58 SGD higher.
- **Capacity:** Larger widths and embeddings did not consistently improve validation RMSE. For example, at  $h = 128$ , increasing  $d$  from 12 to 16 worsened the best RMSE to 65,599.97 SGD.
- **Overfitting evidence:** After epoch 13, training RMSE declined from 31,207.16 to 30,627.12 SGD while validation RMSE rose from 59,766.91 to 69,319.56 SGD by epoch 17, supporting early stopping and indicating overfitting in later epochs.
- **Generalisation:** Retraining ended with training RMSE 38,591.78 SGD and test RMSE 90,595.88 SGD. This gap indicates limited generalisation; changes between the training years and 2022 may also contribute.

## A2

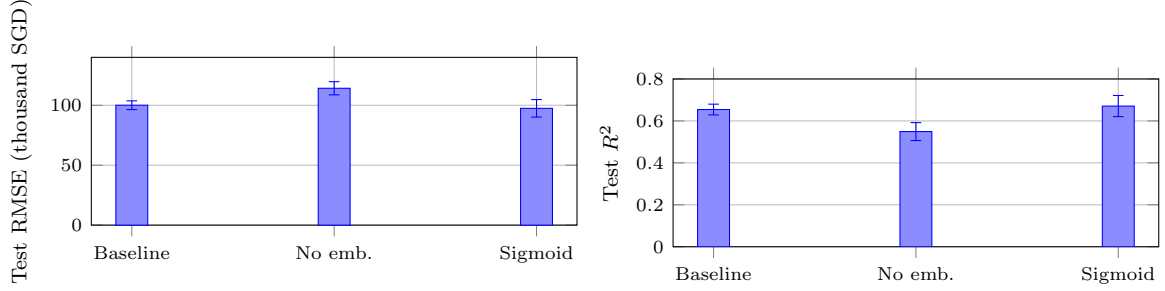
- **Protocol:** All three models were trained from scratch on 2017–2021 for 13 epochs with  $h = 128$ ,  $d = 12$  where applicable, Adam (learning rate 0.01), and batch size 512. Preprocessing was fitted on the combined training set.
- **Seeds and evaluation:** Seeds {42, 43, 44, 45, 46} were used for each model, with identical minibatch orders for matching seeds. Each run was evaluated once on the 2022 test set. Results are mean  $\pm$  sample SD ( $n = 5$ ); error bars show SD. The independent shuffle generator differs from A1(d), so its seed-42 result can differ.

**Per-seed results:** RMSE values are in SGD; all runs used 13 epochs.

| Model              | Seed | Training RMSE | Test RMSE  | Test $R^2$ |
|--------------------|------|---------------|------------|------------|
| Baseline           | 42   | 37,926.73     | 98,124.06  | 0.6679     |
| Baseline           | 43   | 38,258.40     | 97,431.21  | 0.6725     |
| Baseline           | 44   | 37,931.61     | 101,068.42 | 0.6476     |
| Baseline           | 45   | 37,476.46     | 97,601.34  | 0.6714     |
| Baseline           | 46   | 38,618.59     | 106,030.81 | 0.6122     |
| Without embeddings | 42   | 54,406.94     | 114,211.19 | 0.5500     |
| Without embeddings | 43   | 55,308.87     | 117,697.22 | 0.5221     |
| Without embeddings | 44   | 55,730.65     | 119,940.38 | 0.5037     |
| Without embeddings | 45   | 53,933.65     | 113,674.84 | 0.5542     |
| Without embeddings | 46   | 54,487.32     | 105,507.78 | 0.6160     |
| Sigmoid            | 42   | 38,004.36     | 96,090.96  | 0.6815     |
| Sigmoid            | 43   | 39,493.07     | 109,142.47 | 0.5891     |
| Sigmoid            | 44   | 38,728.61     | 89,514.55  | 0.7236     |
| Sigmoid            | 45   | 36,920.54     | 98,480.10  | 0.6654     |
| Sigmoid            | 46   | 38,093.72     | 94,102.55  | 0.6945     |

**Five-seed summary:**

| Model              | Test RMSE (SGD)           | Test $R^2$          |
|--------------------|---------------------------|---------------------|
| Baseline           | $100,051.17 \pm 3,652.75$ | $0.6543 \pm 0.0256$ |
| Without embeddings | $114,206.28 \pm 5,500.29$ | $0.5492 \pm 0.0427$ |
| Sigmoid            | $97,466.13 \pm 7,310.91$  | $0.6708 \pm 0.0504$ |



### (a) Removing Categorical Embeddings

- **Modification:** All four embedding tables were removed. The first layer changed from  $54 \rightarrow 128$  to  $6 \rightarrow 128$ ; LayerNorm, ReLU, and the output layer were retained. Trainable parameters decreased from 8,649 to 1,281.
- **Result:** Mean test RMSE increased by 14.15%, and mean  $R^2$  decreased from 0.6543 to 0.5492. RMSE worsened for four of five matched seeds.
- **Interpretation:** Categorical inputs provide useful information beyond the continuous features. Mean training RMSE also rose from 38,042.36 to 54,773.49 SGD, consistent with reduced fitting ability. This ablation removes both information and parameters, so their effects cannot be separated.

### (b) Replacing ReLU with Sigmoid

- **Modification:** Only the hidden activation changed to Sigmoid.
- **Result:** Mean test RMSE decreased by 2.58%, and mean  $R^2$  increased to 0.6708. However, RMSE SD increased from 3,652.75 to 7,310.91 SGD; Sigmoid improved RMSE for three of five matched seeds.
- **Interpretation:** Sigmoid changes the bounded hidden representation and gradient behaviour while leaving capacity unchanged in parameter count. Mean training RMSE was similar (38,248.06 versus 38,042.36 SGD). Its modest average test improvement and larger seed variation do not establish a reliable advantage over ReLU in this experiment.

## A3

### (a) Selecting $\lambda$

- **Architecture:** Both branches share the 54-dimensional input. The deep branch retains the A1 MLP ( $h = 128$ ,  $d = 12$ ); the wide branch is a single  $54 \rightarrow 1$  linear layer. Outputs are combined as  $\text{MLP}(x) + \lambda \text{Linear}(x)$  before inverse target scaling.
- **Search protocol:**  $\lambda \in \{0, 0.25, 0.5, 1, 2\}$ ; seed 42; training on 2017–2020 and validation on 2021. Adam, learning rate 0.01, batch size 512, maximum 20 epochs, and early-stopping patience 4 with minimum improvement  $10^{-4}$  were retained. All candidates used identical initial weights and minibatch orders.
- **Selection criterion:** Lowest validation RMSE achieved during training. Results for all five candidates are shown below.

| $\lambda$ | Best validation RMSE (SGD) | Best epoch | Epochs run |
|-----------|----------------------------|------------|------------|
| 0         | 59,720.34                  | 4          | 8          |
| 0.25      | 60,935.69                  | 4          | 8          |
| 0.5       | 62,760.50                  | 4          | 8          |
| 1         | 62,025.90                  | 6          | 10         |
| 2         | 63,258.46                  | 6          | 10         |

- **Optimal value:**  $\lambda = 0$ , with validation RMSE 59,720.34 SGD at epoch 4; training stopped at epoch 8. The linear branch contributes nothing at this value, so the selected predictor reduces to the baseline. Positive coefficients did not improve validation performance in this search.

**(b) Five-Seed Evaluation**

**(c) Discussion**

**A4**